

Designing Correct Test Cases

30 Practice Problems — If-Else, Loops, Strings, Arrays & Arithmetic

Instructions: For each program described below, design a complete set of test cases. For every test case, write down: (1) the Input, (2) the Expected Output, and (3) a one-line reason for why that case matters (e.g. normal case, boundary value, invalid input, special/edge case). Aim to cover normal cases, boundary values, and edge/invalid cases for each program — a good answer usually has 4 to 7 test cases per question.

Section: If-Else

Q1. Leap Year Checker

A program reads an integer YEAR and prints 'Leap Year' if it is a leap year (divisible by 4, but if divisible by 100 it must also be divisible by 400), otherwise it prints 'Not a Leap Year'.

Q2. Triangle Validity & Type

A program reads three side lengths A, B, C. It prints 'Invalid Triangle' if the sides cannot form a triangle (sum of any two sides must exceed the third). Otherwise it prints 'Equilateral', 'Isosceles', or 'Scalene'.

Q3. Sign of a Number

A program reads a number and prints 'Positive', 'Negative', or 'Zero'.

Q4. Grade Calculator

A program reads marks (0-100) and prints a grade: A for ≥ 90 , B for ≥ 75 and < 90 , C for ≥ 50 and < 75 , D for ≥ 35 and < 50 , F otherwise. Marks outside 0-100 should print 'Invalid Marks'.

Q5. Largest of Three Numbers

A program reads three numbers and prints the largest one. If two or all three are equal, it should still name the correct value(s).

Q6. Ticket Category by Age

A program reads an age and prints 'Child' for $\text{age} < 12$, 'Adult' for $12 \leq \text{age} < 60$, and 'Senior' for $\text{age} \geq 60$. Negative ages are invalid.

Section: Loops

Q7. Factorial of N

A program reads an integer N and prints N! (factorial) using a loop.

Q8. Prime Number Check

A program reads an integer and prints 'Prime' or 'Not Prime' using a loop that checks divisibility.

Q9. Sum of Digits

A program reads an integer (possibly negative) and prints the sum of its digits using a loop.

Q10. Fibonacci Series up to N terms

A program reads N and prints the first N terms of the Fibonacci series using a loop.

Q11. GCD of Two Numbers

A program reads two integers and prints their GCD using a loop (e.g. repeated subtraction or the Euclidean method written as a loop).

Q12. Reverse a Number

A program reads an integer and prints it reversed using a loop (e.g. 1230 -> 321, and negative numbers keep their sign, e.g. -45 -> -54).

Section: Strings

Q13. Palindrome Check

A program reads a string and prints 'Palindrome' or 'Not Palindrome'. Comparison should ignore case.

Q14. Vowel and Consonant Count

A program reads a string and prints the count of vowels and consonants in it, ignoring digits, spaces, and punctuation.

Q15. Anagram Check

A program reads two strings and prints 'Anagram' if one is a rearrangement of the other's letters (ignoring case and spaces), else 'Not Anagram'.

Q16. Remove Whitespace

A program reads a string and prints it with all spaces (leading, trailing, and in between) removed.

Q17. Title Case Conversion

A program reads a sentence and prints it with the first letter of every word capitalized and the rest lowercase.

Q18. Substring Occurrence Count

A program reads a main string and a substring, and prints how many times the substring occurs in the main string, including overlapping occurrences.

Section: Arrays

Q19. Maximum and Minimum in an Array

A program reads an array of integers and prints the maximum and minimum values.

Q20. Reverse an Array In Place

A program reads an array and prints it reversed, using in-place swapping (no extra array).

Q21. Second Largest Element

A program reads an array and prints the second largest distinct value. If no such value exists (e.g. all elements equal, or fewer than 2 elements), it prints 'Not Found'.

Q22. Remove Duplicates from a Sorted Array

A program reads a sorted array and prints it with consecutive duplicate values removed.

Q23. Check if an Array is Sorted

A program reads an array and prints 'Sorted' if it is sorted in non-decreasing order, else 'Not Sorted'.

Q24. Binary Search

A program reads a sorted array and a target value, and prints the index of the target using binary search, or -1 if not found.

Section: Arithmetic

Q25. Simple Calculator

A program reads two numbers and an operator (+, -, *, /) and prints the result. Division by zero must be handled without crashing.

Q26. Average of N Numbers

A program reads N followed by N numbers and prints their average, rounded to 2 decimal places.

Q27. Celsius <-> Fahrenheit Conversion

A program reads a temperature value and a unit ('C' or 'F') and prints the converted value in the other unit.

Q28. Compound Interest Calculator

A program reads principal, annual rate (%), and time (years) and prints the compound interest, compounded annually, rounded to 2 decimals.

Q29. Divisibility Check

A program reads two integers A and B and prints whether A is divisible by B. Division by zero (B = 0) must be handled gracefully.

Q30. Power Function

A program reads a base and an integer exponent (which may be negative or zero) and prints $\text{base}^{\text{exponent}}$ as a float.